

Shongket

Beyond the mobile coverage

A handheld LoRa crew communication system

with an optional Bluetooth phone companion

Course / University: [ADD COURSE AND UNIVERSITY]

Student name(s): [ADD STUDENT NAME(S)]

Professor: [ADD PROFESSOR NAME]

Submission date: [ADD DATE]

Executive summary

Shongket is a prototype for a small crew that needs to exchange messages when mobile coverage or internet service is unavailable. ESP32 handhelds communicate over LoRa radio. A bonded phone can use Bluetooth to view information and compose a message through one handheld.

The central idea

A new radio packet can be passed on by nearby handhelds. Duplicate suppression, a random relay delay, and a three-hop limit help control repeated transmissions. The design has no central server, coordinator, or route table.

What this report covers

- The handheld, phone app, and their roles in the system.
- How a direct or whole-crew message moves through the network.
- What the code and automated tests establish, and what still needs hardware evidence.

Project status

Implemented prototype with automated tests. Field results are not claimed.

Problem and objectives

Conventional phone messaging depends on mobile infrastructure. In places with weak or absent coverage, a small team still needs a way to share short updates, learn who is nearby, and locate crew members when GPS is available.

Design objectives

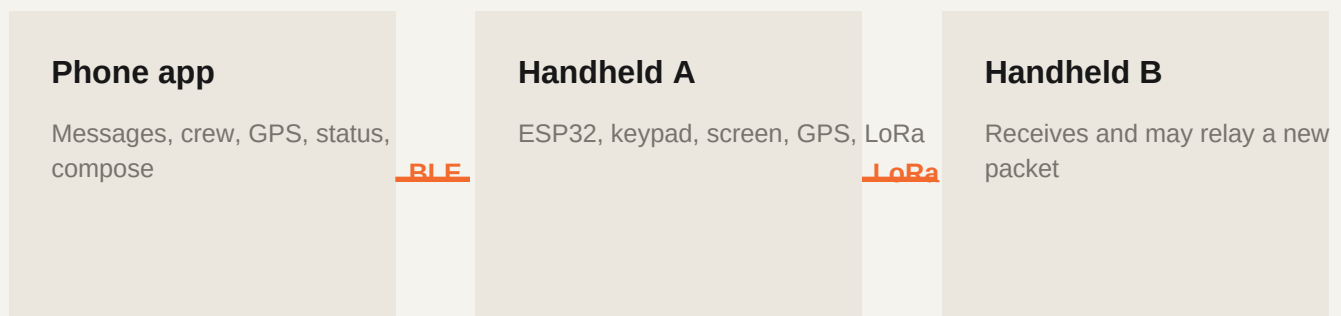
- Let up to 12 handhelds exchange direct and whole-crew messages without a server.
- Allow a message to move through a few neighboring devices when direct radio contact is unavailable.
- Protect radio traffic from outsiders without the crew key.
- Keep basic messaging available on the handheld even when no phone is connected.
- Offer a simple phone view of messages, crew presence, location, and radio status.

Prototype boundary

Shongket is intended for short crew messages, not emergency dispatch. SOS, cloud backup, accounts, and automatic crew-key rotation are not implemented. Real-world range and reliability require controlled hardware trials.

System overview

Each person can use the handheld alone. A nearby bonded phone adds a larger interface but does not become a radio node.



Hardware in each handheld

The design combines an ESP32 controller, SX1278 LoRa radio, ST7789 display, keypad, GPS receiver, and buzzer. The device keeps a small message history and peer directory in RAM.

Communication paths

- LoRa carries encrypted crew traffic between handhelds.
- Bluetooth connects one bonded phone to a nearby handheld.
- GPS supplies location information when a usable fix is available.

How a message moves

1 Compose

Write a short direct message on the device or paired phone.

2 Protect and transmit

The handheld encrypts and authenticates a radio packet using the shared crew key.

3 Check and relay

A receiver checks the packet, ignores copies already seen, and may relay a new packet after a random delay.

4 Receive and acknowledge

The intended handheld stores a direct message and returns an authenticated acknowledgement.

Relaying stops after three hops. A sender may retry a direct message, while the recipient avoids storing the same logical message twice.

Handheld and phone experience

Handheld

- Use the keypad and screen to browse messages, compose text, view the crew, and check device status.
- See a north-up radar of reported peer positions when local GPS has a usable fix.
- Read short status labels for radio, GPS, Bluetooth, and message activity.

Mobile app

- Messages: read, compose, and mark items as read.
- Crew: see known members and their online and location states.
- Radio: inspect handheld health, GPS, and diagnostics.
- Connect: bond with a handheld; mock mode can demonstrate the interface without radio hardware.

Important distinction

The handheld remains the source of truth and handles LoRa traffic.

Security and delivery

Radio protection

The implementation uses AES-128-CCM with a shared crew key. It encrypts packet contents and detects tampering. A device without the key cannot read or create valid crew traffic. Direct messages are not private from another trusted device that holds the same shared key.

Meaning of delivery

- Direct message: delivered means the recipient accepted the message and an acknowledgement returned.
- Whole-crew message: delivered means at least one other device accepted and relayed it. It does not prove that every member received it.

Why controlled flooding

A changing small crew can relay messages without calculating routes. This simplifies the network, but extra transmissions may consume airtime. Duplicate checks, bounded queues, random delay, and the hop limit reduce that cost.

Implementation and validation

The repository contains device firmware, the phone app, protocol documentation, and automated tests. This page separates evidence available in the repository from hardware observations that must be added before submission.

Available code-based evidence

- Firmware test code covers packet serialization, relay state, retries, crew data, mobile framing, UI input, and radar math.
- Mobile tests cover BLE framing, request handling, pagination, busy sends, and event sequence gaps.
- The firmware also runs cryptographic self-tests at boot before starting LoRa.

Hardware demonstration evidence

[ADD DEVICE PHOTOS AND OBSERVED RESULTS HERE.] Suggested observations: two-device discovery and direct acknowledgement; three-device relay with return acknowledgement; phone pairing and message view; GPS/radar when a valid fix exists. Record setup, outcome, and date. Do not replace this box with unmeasured range claims.

Limits and next steps

Current limits

- The design targets up to 12 handhelds and at most three relay hops.
- Messages and drafts are kept in RAM; this is not a persistent archive.
- A shared crew key is configured on devices; key rotation is not automated.
- GPS and radar depend on a usable fix; field range has not been documented here.
- SOS is intentionally absent from the current prototype.

Next validation steps

Complete a staged bench trial with one, two, and three handhelds; record direct acknowledgements, relay behavior, phone pairing, and GPS behavior. Then run an outdoor trial with documented conditions and repeatable measurements.

Conclusion

Shongket brings together an independent handheld, a small LoRa crew network, and an optional phone companion. The repository explains and tests the implementation. Hardware evidence should be attached before claims about field performance are made.

Technical references: [device-firmware/PROJECT_BRIEF.md](#), [LC3_PROTOCOL.md](#), [MOBILE_API.md](#), and the mobile-app test suite.